Qns	Answer	Marks
5(a)	When an object freefalls, the **ONLY** force acting on the object is its weight (= mg). Hence, the freefall acceleration $a = \frac{F}{m} = \frac{mg}{m} = g$ is equal to the gravitational field strength g. OR By definition of gravitational field strength, $g = F_{graviational}/m$ and from Newton's 2nd Law, for a freely falling body, the **ONLY** force acting is gravitational force hence $F_{net} = mg = ma$. Therefore, $a = g$.	**M1** **A1**
5(b)	$g = \frac{GM}{r^2}$ $M = \frac{gr^2}{G} = \frac{9.81 \times (6.37 \times 10^6)^2}{6.67 \times 10^{-11}} = 5.9679 \times 10^{24} = 5.97 \times 10^{24}$	**B1**

Qns	Answer	Marks
5(c)	$\phi = -\frac{GM}{r} = -\frac{6.67\times10^{-11}\times5.97\times10^{24}}{6.37\times10^{6}} = -6.25\times10^{7}\ \text{J kg}^{-1}$	B1
5(d)	The gravitational potential is taken to be zero at infinity. Since the gravitational force is attractive, the work done by the external force to bring the test mass from infinity to a point is negative. Hence, the gravitational potential is negative.	B1 B1 A0